MATHEMATICAL ANALYSIS 1
HOMEWORK 3

(1) (a) Let $X = \{1, 2, 3\}$ and $Y = \{a, b\}$.
  (i) Give an example of a function $f : X \to Y$ with $\mathrm{dom}(f) = X$ that is surjective.
  (ii) Explain why no function $f : X \to Y$ with $\mathrm{dom}(f) = X$ can be injective.
  (iii) Give an example of a function $f : X \to Y$ that is injective. What is the largest possible number of elements of $\mathrm{dom}(f)$ for such an f? Can such an f also be surjective?

(b) Determine whether each of the following functions is injective, surjective, or bijective (explain your answers, either using the definitions or by sketching the graph and considering horizontal lines):
  (i) $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$
  (ii) $f : [0, +\infty) \to \mathbb{R}$, $f(x) = x^2$
  (iii) $f : [0, +\infty) \to [0, +\infty)$, $f(x) = x^2$
  (iv) $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^3$

(2) Let $f(x) = \frac{2x+1}{x-3}$, with $\mathrm{dom}(f) = \mathbb{R} \setminus \{3\}$.
(a) Show that f is injective.
(b) Determine $\mathrm{im}(f)$.
(c) Find the inverse function f^{-1}, and state $\mathrm{dom}(f^{-1})$ and $\mathrm{im}(f^{-1})$.
(d) Verify that $f(f^{-1}(y)) = y$ for all $y \in \mathrm{dom}(f^{-1})$, and that $f^{-1}(f(x)) = x$ for all $x \in \mathrm{dom}(f)$.

(3) Prove the following lemma:

Lemma. *Let $f, g : \mathbb{R} \to \mathbb{R}$ be monotonically increasing on some $A \subseteq \mathbb{R}$. Then $f + g$ is also monotonically increasing on A. If either f or g are* strictly *increasing on A, then so is $f + g$. The same statements hold if we replace everywhere the word 'increasing' with the word 'decreasing'.*

(4) Show that if $f : A \subseteq \mathbb{R} \to \mathbb{R}$ is monotone increasing on A, then $-f$ is monotone decreasing on A.

(5) For the following pairs of functions f and g, compute $f \circ g$ and $g \circ f$, and determine their domains:
(a) $f(x) = x^2$, $g(x) = x + 1$
(b) $f(x) = \sqrt{x}$, $g(x) = x - 2$
(c) $f(x) = \frac{1}{x}$, $g(x) = x^2 + 1$
(d) $f(x) = \sin(x)$, $g(x) = 2x$

(6) Prove the following lemma:

Lemma. *Let $f, g : \mathbb{R} \to \mathbb{R}$. Then:*

$$\begin{aligned} f, g \text{ are both monotone increasing} &\quad\Rightarrow\quad g \circ f \text{ is monotone increasing.} \\ f, g \text{ are both monotone decreasing} &\quad\Rightarrow\quad g \circ f \text{ is monotone increasing.} \\ f, g \text{ are monotone of different kinds} &\quad\Rightarrow\quad g \circ f \text{ is monotone decreasing.} \end{aligned}$$

(7) Let t_c and s_c be the translation and scaling functions, respectively. Let r be the reflection function. Consider the function $f(x) = x^3$ restricted to the interval $[-1, 1]$. Sketch the following:
(a) $f \circ t_c$ for $c = -1, 0, 1$
(b) $t_c \circ f$ for $c = -1, 0, 1$
(c) $f \circ s_c$ for $c = \frac{1}{2}, 2$
(d) $s_c \circ f$ for $c = \frac{1}{2}, 2$
(e) The difference $f \circ r - r \circ f$.

(8) Sketch the graphs of the following functions (choose a relevant part of the graph to sketch):
(a) $\sin(x)$, $\sin(2x)$, $\sin(-x)$
(b) $\log_{\frac{1}{5}} x$, $\log_5 x$, $\log_5(x^2)$

(c) $\cos(x)$, $\cos(2x)$, $\cos(x - \frac{\pi}{2})$

(9) For the following functions f and subsets $A \subseteq \mathrm{dom}(f)$, determine $\sup_A f$, $\inf_A f$, and whether the maximum and minimum are attained on A:
   (a) $f(x) = x^2$, $A = [0, 2]$
   (b) $f(x) = x^2$, $A = (0, 2)$
   (c) $f(x) = x^3 + x^2 + 2x + 1$, $A = [0, 2]$
   (d) $f(x) = \frac{1}{x}$, $A = (0, 1]$
   (e) $f(x) = \frac{1}{x}$, $A = [1, +\infty)$
   (f) $f(x) = e^x$, $A = [-1, 1]$
   (g) $f(x) = \frac{x}{1+x}$, $A = [0, +\infty)$
   (h) $f(x) = \sin(x)$, $A = (0, \pi)$
   (i) $f(x) = \arctan(x)$, $A = [0, +\infty)$

   For parts (g)–(i), explain your answers using the definitions of supremum and infimum.

(10) Using trigonometric identities, simplify the following expressions:
   (a) $\frac{\sin(x)}{\cos(x)} + \frac{\cos(x)}{\sin(x)}$
   (b) $(\sin(x) + \cos(x))^2 + (\sin(x) - \cos(x))^2$
   (c) $\frac{1-\cos(2x)}{\sin(2x)}$

(11) Prove the following equalities:
   (a) $\tan(x) + \cot(x) = \frac{2}{\sin(2x)}$
   (b) $\cos(3x) = 4\cos^3(x) - 3\cos(x)$

Additional Practice Problems

⋆⋆ These problems are for you to practice ⋆⋆
⋆⋆ They are not part of the homework assignment ⋆⋆

(1) Sketch the graphs of the following functions (choose a relevant part of the graph to sketch):
   (a) $\sin(x+\frac{\pi}{2})$, $\sin(x)+1$
   (b) $\log_{10}(\frac{2}{x})$, $\ln(-x)$, $\ln(x^{-1})$
   (c) $-\cos(x)$, $\cos(-x)$
   (d) $\tan(x)$, $\tan(2x)$, $\tan(x+\frac{\pi}{4})$, $-\tan(x)$
   (e) $\arcsin(x)$, $\arcsin(2x)$, $\arcsin(-x)$, $\arcsin(x)+\frac{\pi}{2}$
   (f) $\arctan(x)$, $\arctan(2x)$, $\arctan(-x)$, $-\arctan(x)$

(2) Sketch the graphs of the following power functions:
   (a) $x^{1/2}$, $x^{1/3}$, $x^{1/4}$
   (b) x^{-1}, x^{-2}, x^{-3}
   (c) $x^{2/3}$, $x^{3/2}$, $x^{-1/2}$
   (d) $(x-1)^{1/2}$, $(x+2)^{-1}$, $2x^3$

(3) For the following pairs of functions f and g, compute $f\circ g$ and $g\circ f$, and determine their domains:
   (a) $f(x)=e^x$, $g(x)=\ln(x)$
   (b) $f(x)=|x|$, $g(x)=x-3$
   (c) $f(x)=\sqrt{x}$, $g(x)=x^2$
   (d) $f(x)=\frac{1}{x-1}$, $g(x)=\frac{1}{x}$
   (e) $f(x)=x^{1/3}$, $g(x)=x^3$
   (f) $f(x)=x^2+1$, $g(x)=\sqrt{x-1}$

(4) For the following functions f and subsets $A\subseteq \mathrm{dom}(f)$, determine $\sup_A f$, $\inf_A f$, and whether the maximum and minimum are attained on A:
   (a) $f(x)=e^x$, $A=(-1,1)$
   (b) $f(x)=\sqrt{4-x^2}$, $A=(-2,2)$
   (c) $f(x)=\arctan(x)$, $A=\mathbb{R}$
   (d) $f(x)=\frac{1}{1-x}$, $A=[0,1)$
   (e) $f(x)=x(2-x)$, $A=(0,2)$
   (f) $f(x)=x^{1/3}$, $A=[-8,8]$
   (g) $f(x)=x^{-2}$, $A=[1,+\infty)$

(5) Determine whether the following functions are even, odd, or neither:
   (a) $f(x)=x^4+x^2$
   (b) $f(x)=x^5-x^3$
   (c) $f(x)=x^3+x^2$
   (d) $f(x)=\frac{x}{x^2+1}$
   (e) $f(x)=x^{-2}$
   (f) $f(x)=|x|+x$

(6) Using trigonometric identities, simplify the following expressions:
   (a) $\sin(x+y)\sin(x-y)$
   (b) $\cos^4(x)-\sin^4(x)$
   (c) $\frac{1+\tan^2(x)}{1+\cot^2(x)}$

(7) Prove the following equality:
   (a) $\frac{\sin(x)+\sin(3x)}{\cos(x)+\cos(3x)}=\tan(2x)$

(8) Simplify the following expressions involving powers:
   (a) $(x^3)^{1/2}\cdot x^{-1/2}$
   (b) $\frac{x^{2/3}\cdot x^{1/2}}{x^{-1/6}}$
   (c) $(x^{-2}y^3)^{-1/2}$
   (d) $\left(\frac{x^{1/3}}{x^{-2/3}}\right)^3$

Homework 3 Solutions

(1) (a) (i) For example $f(1) = a$, $f(2) = b$, $f(3) = b$. Then $\mathrm{im}(f) = \{a, b\} = Y$, so f is surjective.

(ii) If $\mathrm{dom}(f) = X$, the three elements $1, 2, 3$ are sent to only two possible values, so at least two of them have the same image, i.e. some $y \in Y$ has at least two pre-images. Hence f is not injective.

(iii) Recall that $\mathrm{dom}(f)$ may be a proper subset of X. For example, let $\mathrm{dom}(f) = \{1, 2\}$ with $f(1) = a$, $f(2) = b$; then every $y \in Y$ has exactly one pre-image, so f is injective. By the argument of (ii), an injective f cannot have three elements in its domain, so the largest possible number of elements of $\mathrm{dom}(f)$ is 2. The example above is also surjective, since $\mathrm{im}(f) = \{a, b\} = Y$; thus it is a bijection between $\{1, 2\}$ and Y.

(b) (i) Not injective: $f(-1) = f(1) = 1$ (the horizontal line $y = 1$ meets the graph twice). Not surjective: $f(x) = x^2 \geq 0$, so $-1 \notin \mathrm{im}(f)$ (the line $y = -1$ does not meet the graph). Hence not bijective.

(ii) Injective: if $x_1, x_2 \geq 0$ and $x_1^2 = x_2^2$, then $(x_1 - x_2)(x_1 + x_2) = 0$; either $x_1 = x_2$, or $x_1 + x_2 = 0$, which for $x_1, x_2 \geq 0$ forces $x_1 = x_2 = 0$. Not surjective, as in (i): $-1 \notin \mathrm{im}(f)$. Hence not bijective.

(iii) Injective, as in (ii). Surjective: for any $y \geq 0$, the number $x = \sqrt{y} \in [0, +\infty)$ satisfies $f(x) = y$. Hence f is bijective. Note that (i)–(iii) all have the same formula; only the domain and codomain change.

(iv) Injective: if $x_1^3 = x_2^3$, then $(x_1 - x_2)(x_1^2 + x_1x_2 + x_2^2) = 0$. Since $x_1^2 + x_1x_2 + x_2^2 = \left(x_1 + \frac{x_2}{2}\right)^2 + \frac{3}{4}x_2^2$, this factor vanishes only if $x_1 = x_2 = 0$. In either case $x_1 = x_2$. Surjective: for any $y \in \mathbb{R}$, the number $x = \sqrt[3]{y}$ satisfies $f(x) = y$. Hence f is bijective.

(2) (a) Let $x_1, x_2 \neq 3$ with $f(x_1) = f(x_2)$. Then $(2x_1 + 1)(x_2 - 3) = (2x_2 + 1)(x_1 - 3)$, i.e. $2x_1x_2 - 6x_1 + x_2 - 3 = 2x_1x_2 - 6x_2 + x_1 - 3$, i.e. $7x_2 = 7x_1$. Hence $x_1 = x_2$, so f is injective.

(b) Let $y \in \mathbb{R}$. We look for $x \neq 3$ with $\frac{2x+1}{x-3} = y$, i.e. $2x + 1 = y(x - 3)$, i.e. $x(y - 2) = 3y + 1$. If $y \neq 2$, this has the unique solution $x = \frac{3y+1}{y-2}$, and $x \neq 3$ (otherwise $3y + 1 = 3y - 6$, which is impossible). If $y = 2$, the equation reads $0 = 7$, which has no solution. Hence $\mathrm{im}(f) = \mathbb{R} \setminus \{2\}$.

(c) By (b), $f^{-1}(y) = \frac{3y+1}{y-2}$, with $\mathrm{dom}(f^{-1}) = \mathrm{im}(f) = \mathbb{R} \setminus \{2\}$ and $\mathrm{im}(f^{-1}) = \mathrm{dom}(f) = \mathbb{R} \setminus \{3\}$.

(d) For $y \neq 2$:

$$f(f^{-1}(y)) = \frac{2 \cdot \frac{3y+1}{y-2} + 1}{\frac{3y+1}{y-2} - 3} = \frac{\frac{6y+2+y-2}{y-2}}{\frac{3y+1-3y+6}{y-2}} = \frac{7y}{7} = y.$$

For $x \neq 3$:

$$f^{-1}(f(x)) = \frac{3 \cdot \frac{2x+1}{x-3} + 1}{\frac{2x+1}{x-3} - 2} = \frac{\frac{6x+3+x-3}{x-3}}{\frac{2x+1-2x+6}{x-3}} = \frac{7x}{7} = x.$$

(3) *Proof.* Let $x, y \in A$ with $x < y$. Since f and g are monotonically increasing:

$$f(x) \leq f(y)$$
$$g(x) \leq g(y)$$

Adding: $(f + g)(x) = f(x) + g(x) \leq f(y) + g(y) = (f + g)(y)$.

If either is strictly increasing, the corresponding inequality is strict, so $f + g$ is strictly increasing. The decreasing case follows similarly. □

(4) *Proof.* Let $x, y \in A$ with $x < y$. Since f is increasing:

$$f(x) \leq f(y) \Rightarrow -f(x) \geq -f(y)$$

So $-f$ is decreasing. □

(5) (a) $f \circ g(x) = (x + 1)^2$, $\mathrm{dom} = \mathbb{R}$
$g \circ f(x) = x^2 + 1$, $\mathrm{dom} = \mathbb{R}$

(b) $f \circ g(x) = \sqrt{x - 2}$, $\mathrm{dom} = [2, \infty)$
$g \circ f(x) = \sqrt{x} - 2$, $\mathrm{dom} = [0, \infty)$

(c) $f \circ g(x) = \frac{1}{x^2+1}$, $\mathrm{dom} = \mathbb{R}$
$g \circ f(x) = \frac{1}{x^2} + 1$, $\mathrm{dom} = \mathbb{R} \setminus \{0\}$

(d) $f \circ g(x) = \sin(2x)$, $\text{dom} = \mathbb{R}$
$g \circ f(x) = 2\sin(x)$, $\text{dom} = \mathbb{R}$

(6) *Proof.* Let $x < y$.

Case 1: Both increasing

$$\begin{aligned} f(x) &\le f(y) \\ g(f(x)) &\le g(f(y)) \quad \text{(since } g \text{ increasing)} \end{aligned}$$

So $g \circ f$ is increasing.

Case 2: Both decreasing

$$\begin{aligned} f(x) &\ge f(y) \\ g(f(x)) &\le g(f(y)) \quad \text{(since } g \text{ decreasing)} \end{aligned}$$

So $g \circ f$ is increasing.

Case 3: Different types. There are two subcases.

(3a) f increasing, g decreasing:

$$\begin{aligned} f(x) &\le f(y) \quad \text{(since } f \text{ increasing)} \\ g(f(x)) &\ge g(f(y)) \quad \text{(since } g \text{ decreasing)} \end{aligned}$$

So $g \circ f$ is decreasing.

(3b) f decreasing, g increasing:

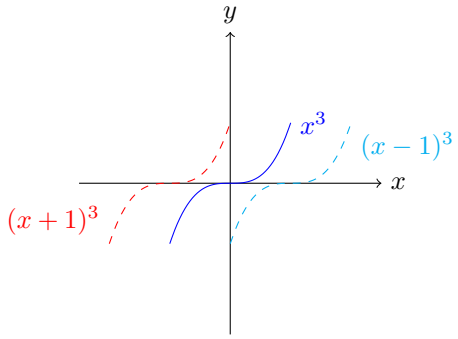
$$\begin{aligned} f(x) &\ge f(y) \quad \text{(since } f \text{ decreasing)} \\ g(f(x)) &\ge g(f(y)) \quad \text{(since } g \text{ increasing)} \end{aligned}$$

So $g \circ f$ is decreasing. □

(7) Here f is restricted to $[-1, 1]$, so each composition is drawn only on its own domain: $\text{dom}(f \circ t_c) = \{x : x + c \in [-1, 1]\} = [-1 - c, 1 - c]$, and $\text{dom}(f \circ s_c) = \{x : cx \in [-1, 1]\} = [-\frac{1}{c}, \frac{1}{c}]$; the compositions $t_c \circ f$ and $s_c \circ f$ keep the domain $[-1, 1]$. (The sketches use the same scale on both axes.)

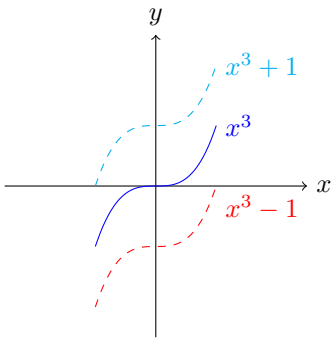
(a) $f \circ t_c$ for $c = -1, 0, 1$:

- $c = -1$: $f \circ t_{-1}(x) = f(x - 1) = (x - 1)^3$
- $c = 0$: $f \circ t_0(x) = f(x) = x^3$
- $c = 1$: $f \circ t_1(x) = f(x + 1) = (x + 1)^3$



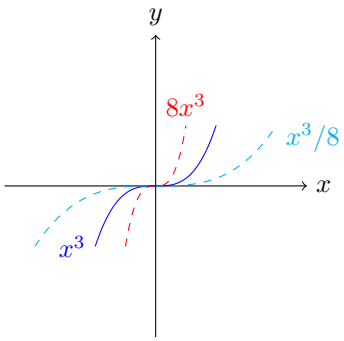
(b) $t_c \circ f$ for $c = -1, 0, 1$:

- $c = -1$: $t_{-1} \circ f(x) = f(x) - 1 = x^3 - 1$
- $c = 0$: $t_0 \circ f(x) = f(x) = x^3$
- $c = 1$: $t_1 \circ f(x) = f(x) + 1 = x^3 + 1$



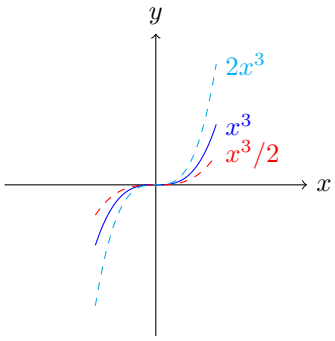
(c) $f \circ s_c$ for $c = \frac{1}{2}, 2$:

- $c = \frac{1}{2}$: $f \circ s_{1/2}(x) = f(x/2) = (x/2)^3 = x^3/8$
- $c = 2$: $f \circ s_2(x) = f(2x) = (2x)^3 = 8x^3$



(d) $s_c \circ f$ for $c = \frac{1}{2}, 2$:

- $c = \frac{1}{2}$: $s_{1/2} \circ f(x) = \frac{1}{2}f(x) = \frac{1}{2}x^3 = x^3/2$
- $c = 2$: $s_2 \circ f(x) = 2f(x) = 2x^3$



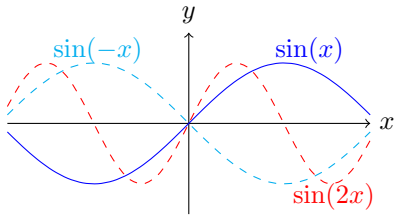
(e) $(f \circ r)(x) = f(-x) = (-x)^3 = -x^3$ and $(r \circ f)(x) = -f(x) = -x^3$, for $x \in [-1, 1]$. Hence

$$(f \circ r - r \circ f)(x) = -x^3 - (-x^3) = 0 \qquad \text{for all } x \in [-1, 1],$$

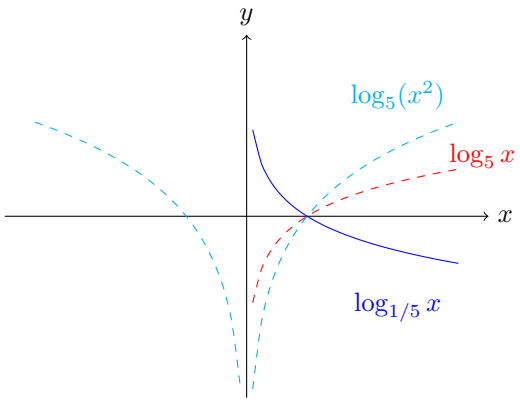
i.e. the difference is the zero function. This happens because f is odd: $f(-x) = -f(x)$.

(8)

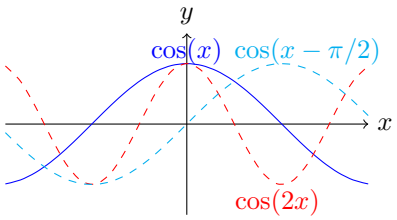
(a)



(b)



(c)



(9) (a) Since x^2 is strictly increasing on $[0,+\infty)$, for $x \in [0,2]$ we have $0 = f(0) \le f(x) \le f(2) = 4$. Both bounds are attained: $\sup_A f = \max_A f = 4$ (at $x = 2$) and $\inf_A f = \min_A f = 0$ (at $x = 0$).

(b) As in (a), since the endpoints are excluded and x^2 is strictly increasing on $[0,+\infty)$, we have $0 < f(x) < 4$ for all $x \in (0,2)$. So 4 is an upper bound and 0 is a lower bound, and neither is attained. They are the best bounds: given $r < 4$, pick $x \in (\sqrt{\max\{r,0\}}, 2)$; then $f(x) = x^2 > r$. Given $r > 0$, pick $x \in (0, \min\{\sqrt{r}, 2\})$; then $f(x) = x^2 < r$. Hence $\sup_A f = 4$ and $\inf_A f = 0$, and there is neither a maximum nor a minimum.

(c) On $[0,+\infty)$ the functions x^3, x^2 and $2x$ are strictly increasing and 1 is constant, so by Lemma 4.2 their sum f is strictly increasing on $[0,2]$. Hence $\inf_A f = \min_A f = f(0) = 1$ and $\sup_A f = \max_A f = f(2) = 8 + 4 + 4 + 1 = 17$.

(d) Since $\frac{1}{x}$ is strictly decreasing on $(0,+\infty)$, we have $f(x) \ge f(1) = 1$ for all $x \in (0,1]$, so $\inf_A f = \min_A f = 1$, attained at $x = 1$. The function is not bounded from above on A: given $M > 0$, the point $x = \frac{1}{M+1} \in (0,1]$ satisfies $f(x) = M + 1 > M$. Hence $\sup_A f = +\infty$, and there is no maximum.

(e) Since $\frac{1}{x}$ is strictly decreasing, $f(x) \le f(1) = 1$ for all $x \in [1,+\infty)$, so $\sup_A f = \max_A f = 1$, attained at $x = 1$. Since $f(x) > 0$ on A, 0 is a lower bound, and it is not attained. It is the greatest lower bound: given $r > 0$, the point $x = 1 + \frac{1}{r} \in A$ satisfies $f(x) = \frac{1}{1+1/r} < \frac{1}{1/r} = r$. Hence $\inf_A f = 0$, and there is no minimum.

(f) Since $e > 1$, the function e^x is strictly increasing, so $e^{-1} = f(-1) \le f(x) \le f(1) = e$ for all $x \in [-1,1]$. Hence $\sup_A f = \max_A f = e$ (at $x = 1$) and $\inf_A f = \min_A f = e^{-1}$ (at $x = -1$).

(g) Write $f(x) = 1 - \frac{1}{1+x}$. Since $\frac{1}{1+x}$ is strictly decreasing on $[0,+\infty)$, f is strictly increasing there, so $\inf_A f = \min_A f = f(0) = 0$. Moreover $f(x) < 1$ for all $x \in A$, so 1 is an upper bound. It is the least upper bound: given $r < 1$, choose $x \ge 0$ with $x > \frac{1}{1-r} - 1$; then $\frac{1}{1+x} < 1 - r$, i.e. $f(x) > r$. Hence $\sup_A f = 1$, and it is not attained (no maximum).

(h) For $x \in (0,\pi)$ we have $0 < \sin(x) \le 1$, with $\sin(\frac{\pi}{2}) = 1$; hence $\sup_A f = \max_A f = 1$, attained at $x = \frac{\pi}{2}$. Since $\sin(x) > 0$ on A, 0 is a lower bound. It is the greatest lower bound: given $r > 0$ (we may assume $r \le 1$), let $x = \arcsin(\frac{r}{2}) \in (0, \frac{\pi}{2}) \subseteq A$; then $\sin(x) = \frac{r}{2} < r$. Hence $\inf_A f = 0$, and it is not attained (no minimum).

(i) The function arctan is strictly increasing (it is the inverse of the strictly increasing function tan on $(-\frac{\pi}{2}, \frac{\pi}{2})$) and $\arctan(0) = 0$, so $\inf_A f = \min_A f = 0$, attained at $x = 0$. Since $\mathrm{im}(\arctan) = (-\frac{\pi}{2}, \frac{\pi}{2})$, we have $f(x) < \frac{\pi}{2}$ for all x, so $\frac{\pi}{2}$ is an upper bound. It is the least upper

bound: given $r < \frac{\pi}{2}$, pick $t \in (\max\{r, 0\}, \frac{\pi}{2})$ and let $x = \tan(t) \geq 0$; then $f(x) = t > r$. Hence $\sup_A f = \frac{\pi}{2}$, and it is not attained (no maximum).

(10) (a) $\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = \tan x + \cot x = \frac{\sin^2 x + \cos^2 x}{\sin x \cos x} = \frac{1}{\sin x \cos x} = \frac{2}{\sin 2x}$

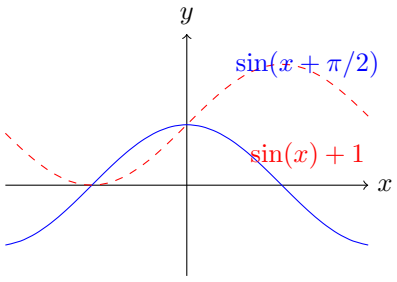
(b) $(\sin x + \cos x)^2 + (\sin x - \cos x)^2 = (1 + 2\sin x \cos x) + (1 - 2\sin x \cos x) = 2$

(c) $\frac{1 - \cos 2x}{\sin 2x} = \frac{2\sin^2 x}{2\sin x \cos x} = \frac{\sin x}{\cos x} = \tan x$

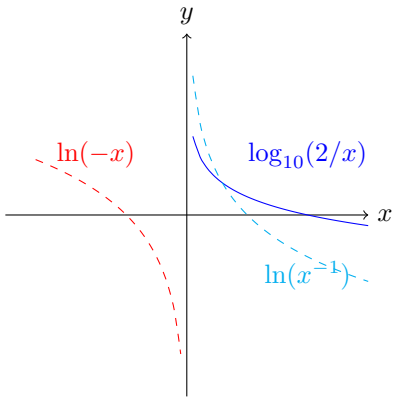
(11) (a) $\tan x + \cot x = \frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = \frac{\sin^2 x + \cos^2 x}{\sin x \cos x} = \frac{1}{\sin x \cos x} = \frac{2}{\sin 2x}$

(b) $\cos 3x = \cos(2x + x) = \cos 2x \cos x - \sin 2x \sin x$
$= (2\cos^2 x - 1)\cos x - 2\sin^2 x \cos x$
$= 2\cos^3 x - \cos x - 2(1 - \cos^2 x)\cos x$
$= 2\cos^3 x - \cos x - 2\cos x + 2\cos^3 x = 4\cos^3 x - 3\cos x$

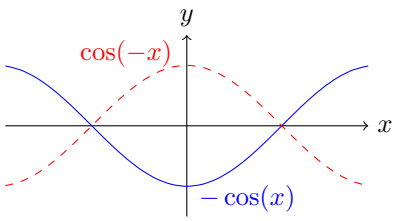
(1) (a)

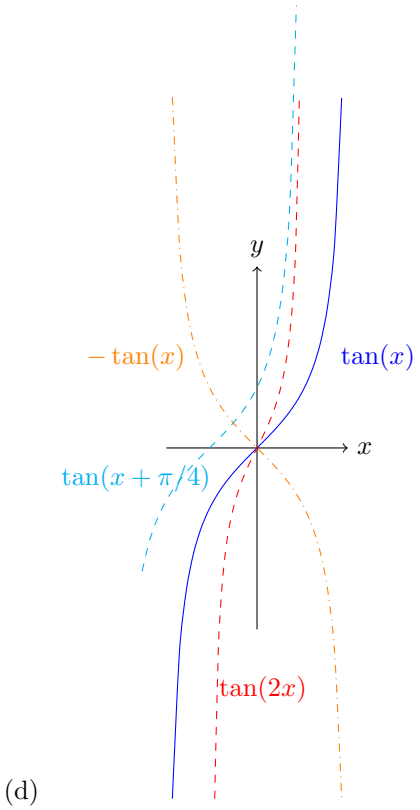


(b)

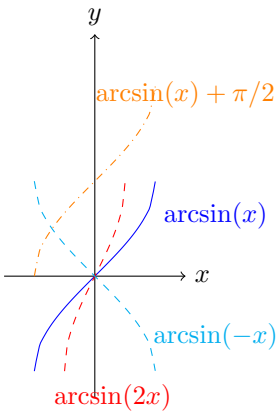


(c)

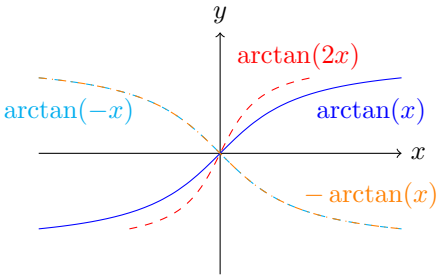



y
x
− tan(x)
tan(x)
tan(x + π/4)
tan(2x)

(d)

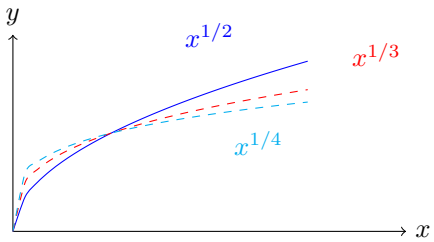

y
x
arcsin(x) + π/2
arcsin(x)
arcsin(−x)
arcsin(2x)

(e)

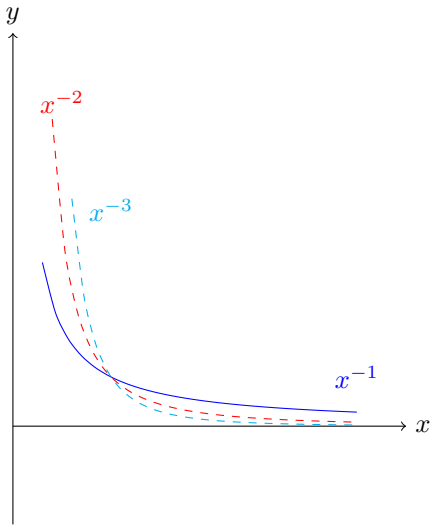

y
x
arctan(2x)
arctan(−x)
arctan(x)
− arctan(x)

(f)

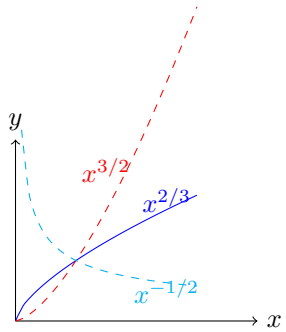
(2) (a)



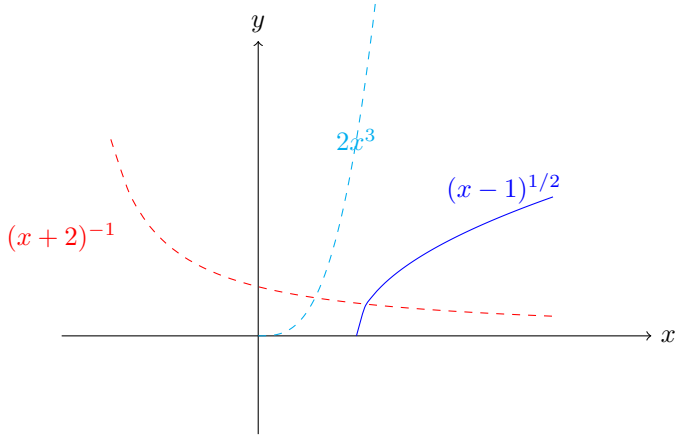
(b)



(c)



(d)



(3) (a) $f \circ g(x) = e^{\ln x} = x$, dom $= (0, +\infty)$
$g \circ f(x) = \ln(e^x) = x$, dom $= \mathbb{R}$
(b) $f \circ g(x) = |x - 3|$, dom $= \mathbb{R}$
$g \circ f(x) = |x| - 3$, dom $= \mathbb{R}$
(c) $f \circ g(x) = \sqrt{x^2} = |x|$, dom $= \mathbb{R}$
$g \circ f(x) = (\sqrt{x})^2 = x$, dom $= [0, +\infty)$
(d) $f \circ g(x) = \frac{1}{\frac{1}{x}-1} = \frac{x}{1-x}$, dom $= \mathbb{R} \setminus \{0, 1\}$
$g \circ f(x) = \frac{1}{\frac{1}{x-1}} = x - 1$, dom $= \mathbb{R} \setminus \{1\}$
(e) $f \circ g(x) = (x^3)^{1/3} = x$, dom $= \mathbb{R}$
$g \circ f(x) = (x^{1/3})^3 = x$, dom $= \mathbb{R}$
(f) $f \circ g(x) = (\sqrt{x-1})^2 + 1 = x$, dom $= [1, +\infty)$
$g \circ f(x) = \sqrt{(x^2+1)-1} = |x|$, dom $= \mathbb{R}$

(4) (a) sup $= e$, inf $= e^{-1}$, neither attained
(b) sup $= 2$, inf $= 0$, neither attained
(c) sup $= \pi/2$, inf $= -\pi/2$, neither attained
(d) sup $= \infty$, inf $= 1$, min attained at $x = 0$, no max
(e) sup $= 1$, inf $= 0$, neither attained
(f) sup $= 2$, inf $= -2$, both attained
(g) sup $= 1$, inf $= 0$, max attained at $x = 1$, no min

(5) (a) Even: $f(-x) = (-x)^4 + (-x)^2 = x^4 + x^2 = f(x)$
(b) Odd: $f(-x) = (-x)^5 - (-x)^3 = -x^5 + x^3 = -f(x)$
(c) Neither: $f(-x) = (-x)^3 + (-x)^2 = -x^3 + x^2 \neq \pm f(x)$
(d) Odd: $f(-x) = \frac{-x}{(-x)^2+1} = \frac{-x}{x^2+1} = -f(x)$
(e) Even: $f(-x) = (-x)^{-2} = x^{-2} = f(x)$
(f) Neither: $f(-x) = |-x| + (-x) = |x| - x \neq \pm f(x)$

(6) (a) $\sin(x+y)\sin(x-y) = \frac{1}{2}[\cos(2y) - \cos(2x)] = \sin^2 y - \sin^2 x$
(b) $\cos^4 x - \sin^4 x = (\cos^2 x - \sin^2 x)(\cos^2 x + \sin^2 x) = \cos 2x$
(c) $\frac{1+\tan^2 x}{1+\cot^2 x} = \frac{\sec^2 x}{\csc^2 x} = \frac{\sin^2 x}{\cos^2 x} = \tan^2 x$

(7) *Proof.* Using sum-to-product identities:

$$\begin{aligned} \frac{\sin x + \sin 3x}{\cos x + \cos 3x} &= \frac{2\sin(2x)\cos x}{2\cos(2x)\cos x} \\ &= \frac{\sin 2x}{\cos 2x} = \tan 2x \end{aligned}$$

□

(8) (a) $(x^3)^{1/2} \cdot x^{-1/2} = x^{3/2} \cdot x^{-1/2} = x^1 = x$
(b) $\frac{x^{2/3} \cdot x^{1/2}}{x^{-1/6}} = x^{2/3+1/2-(-1/6)} = x^{2/3+1/2+1/6} = x^{4/3}$
(c) $(x^{-2}y^3)^{-1/2} = x^1 y^{-3/2} = \frac{x}{y^{3/2}}$
(d) $\left(\frac{x^{1/3}}{x^{-2/3}}\right)^3 = (x^{1/3-(-2/3)})^3 = (x^1)^3 = x^3$